Computational PDEs: Classical Approaches and Physics-Informed Learning

Day 4 — Schottky–Klein primes, close circles, and PINNs

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Book to read

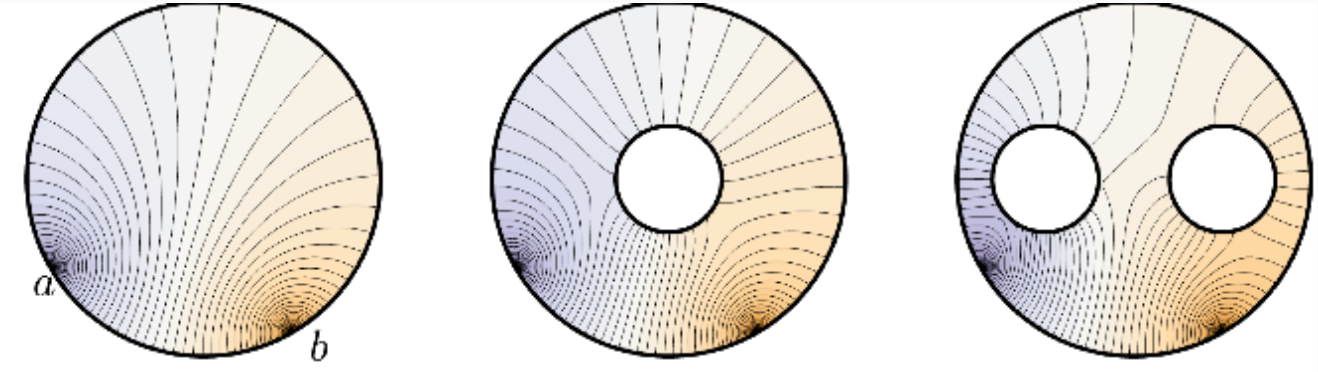
- **Complex variables** by Ablowitz and Fokas
- **Solving Problems in Multiply Connected Domains** by D. G. Crowdy
- **Conformal mapping** by Zeev Nehari

Japanese Sake



Wednesday in one picture

- Simply connected $\to$ unit disk (Tuesday).
- One hole $\to$ **annulus** $D_\zeta = \{\rho < |\zeta| < 1\}$.
- Two or more holes $\to$ unit disk with circular holes (circular canonical domain).



Annulus and the product P

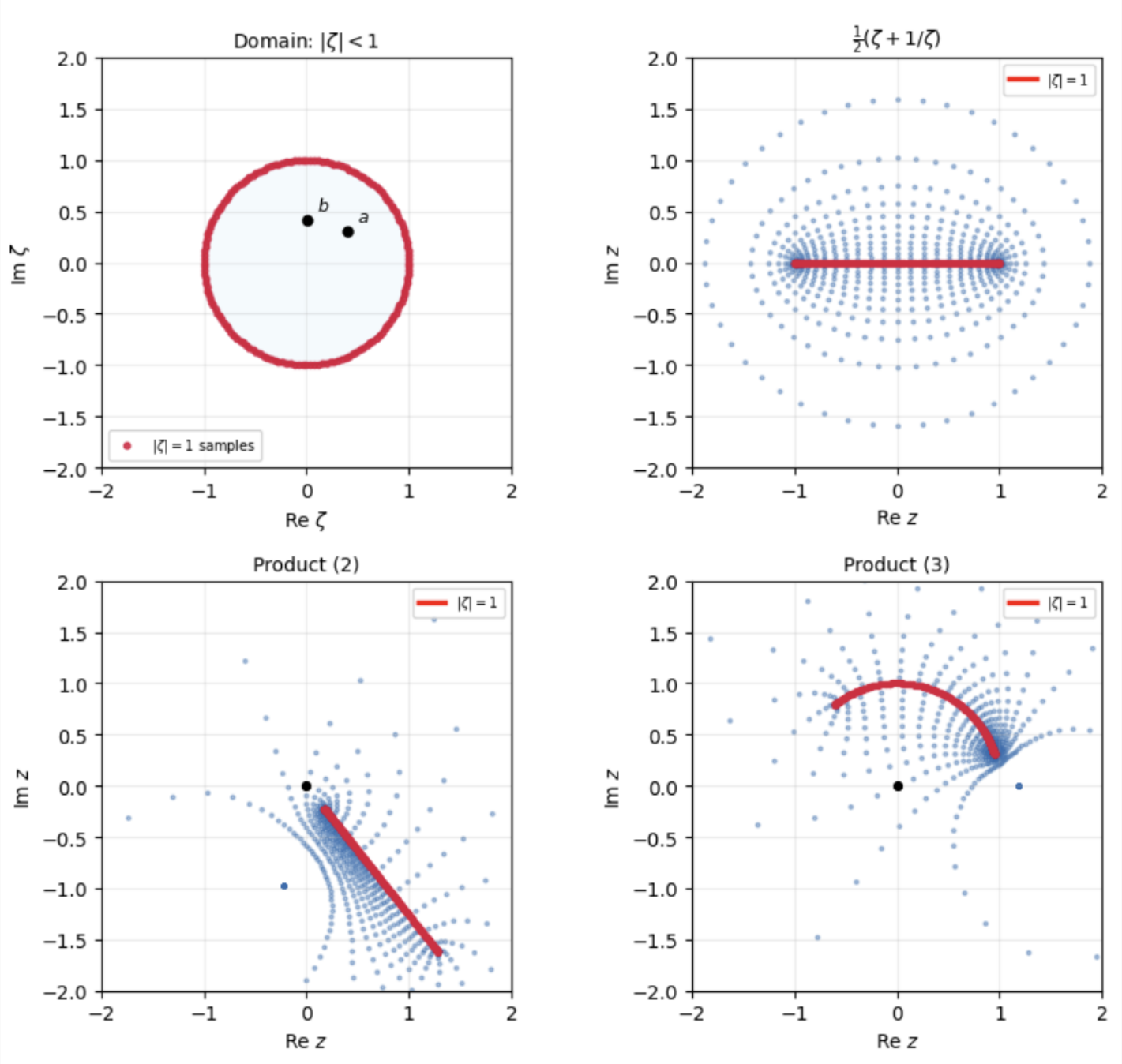


Figure 1: 03_day3_conformal_map_exercises.ipynb

The function $P(\zeta, \rho)$

$$P(\zeta, \rho) := (1 - \zeta) \prod_{n=1}^{\infty} (1 - \rho^{2n}\zeta)\,(1 - \rho^{2n}\zeta^{-1}). \tag{1}$$

Two identities

$$P(\zeta^{-1}, \rho) = -\zeta^{-1}\, P(\zeta, \rho) \qquad \text{(inversion)},$$
$$P(\rho^2\zeta, \rho) = -\zeta^{-1}\, P(\zeta, \rho) \qquad \text{(quasi-periodicity)}.$$

Unique zero in the annulus at $\zeta = 1$. Truncate the product at $N_{\max}$ for numerics. Those two relations become the PINN loss later this morning.

Short exercise

$$P(\zeta, \rho) \coloneqq (1-\zeta) \prod_{n=1}^{\infty} (1-\rho^{2n}\zeta)\,(1-\rho^{2n}\zeta^{-1}). \tag{2}$$

- Prove that

$$P(\rho^2\zeta, \rho) = -\zeta^{-1} P(\zeta, \rho) \tag{3}$$

- Define

$$g(\zeta) = -\frac{P(\zeta, \rho)}{P(-\zeta, \rho)}. \tag{4}$$

  Prove that on $\zeta \in C_0$,

$$\mathrm{Re}[g(\zeta)] = 0. \tag{5}$$

Mapping

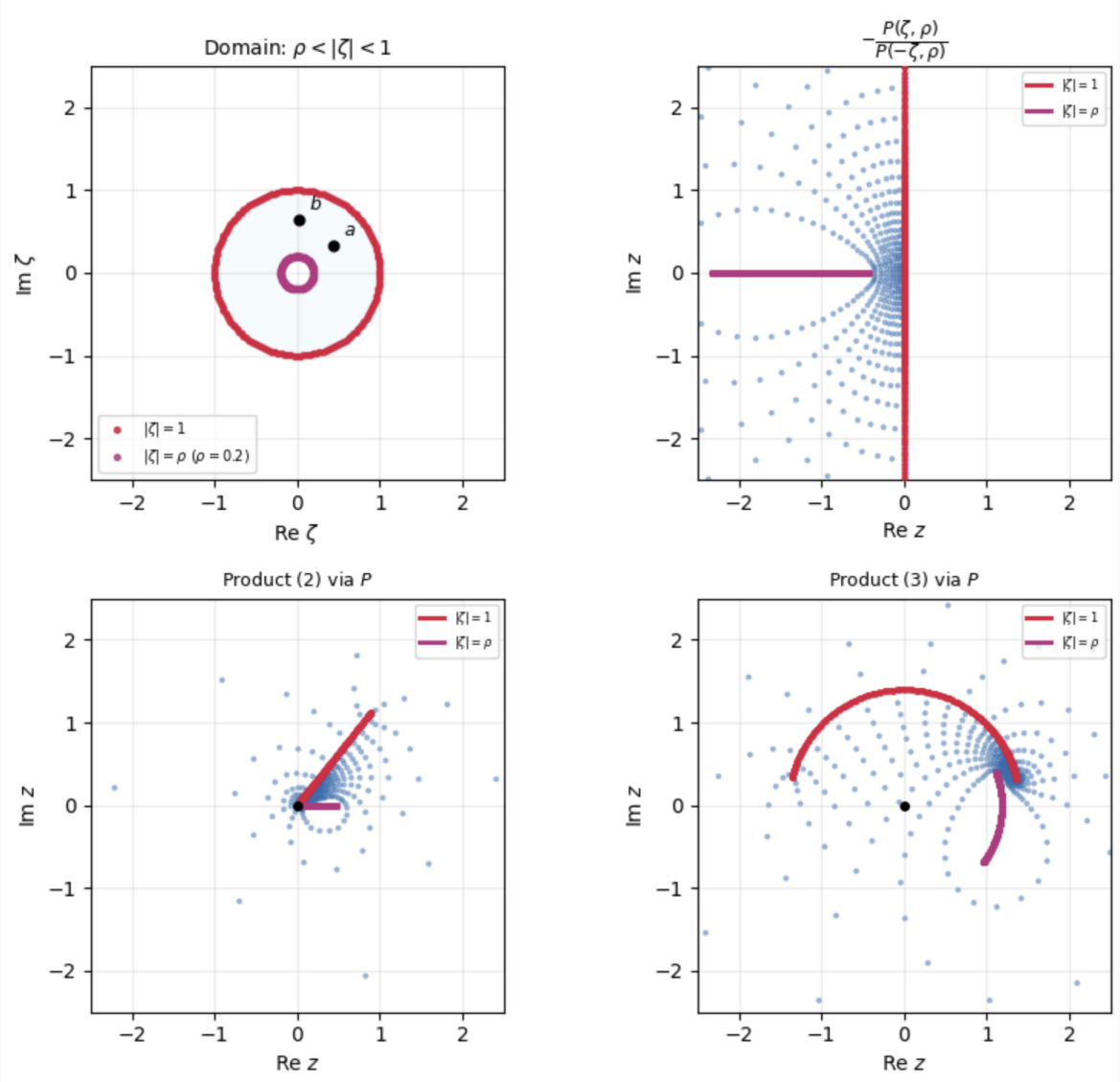


Figure 2: `03_schottky_klein_annulus.ipynb`

Exercise (day3)

Define the cross-ratio:

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}. \tag{6}$$

and define a Möbius transformation:

$$\phi(z) = \frac{az + b}{cz + d}. \tag{7}$$

Prove that

$$(z_1, z_2; z_3, z_4) = (\phi(z_1), \phi(z_2); \phi(z_3), \phi(z_4)). \tag{8}$$

This means that the cross-ratio is invariant under Möbius transformations.

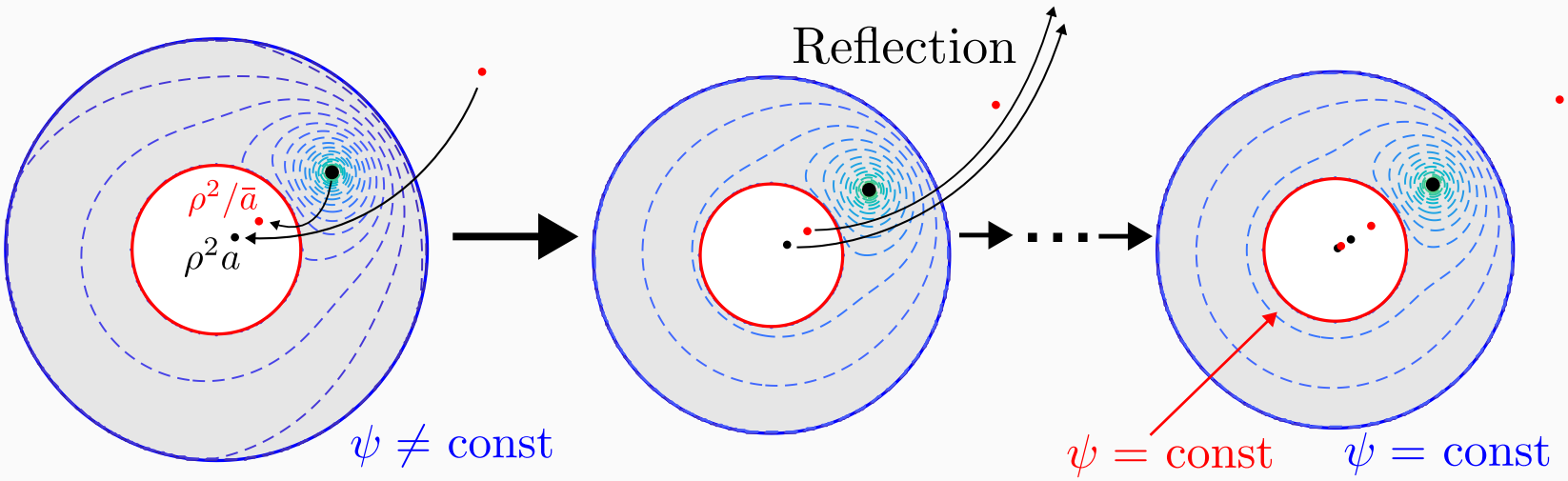
(Advanced) Several holes

From the annulus to several holes

Wednesday built P by inverting in two circles. One more hole is another circle, not another factor ρ^{2n}.

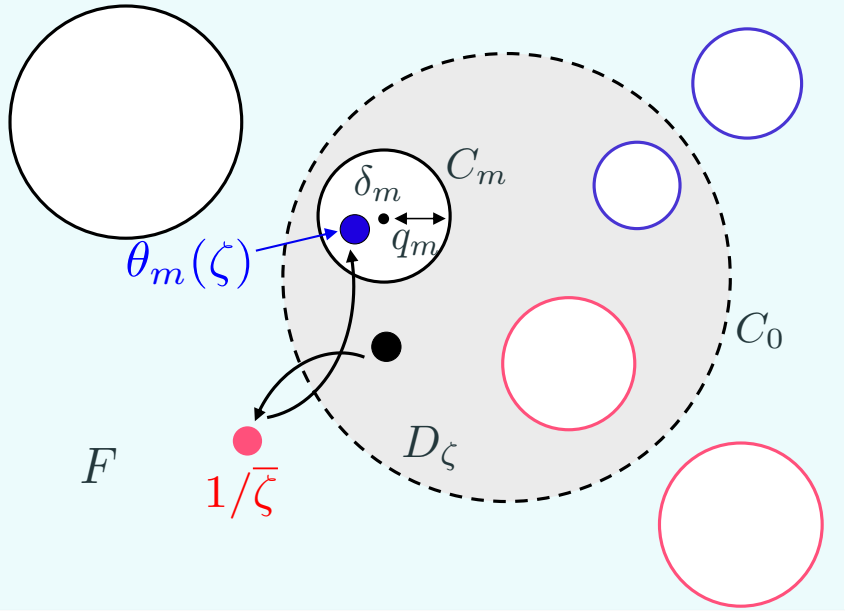
The signed images on the annulus were

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \Big(\log(\zeta - a) - \log(\zeta - 1/\overline{a}) - \log(\zeta - \rho^2/\overline{a}) + \log(\zeta - \rho^2 a) + \log(\zeta - a/\rho^2) - \cdots \Big).$$



Reflection of the circle

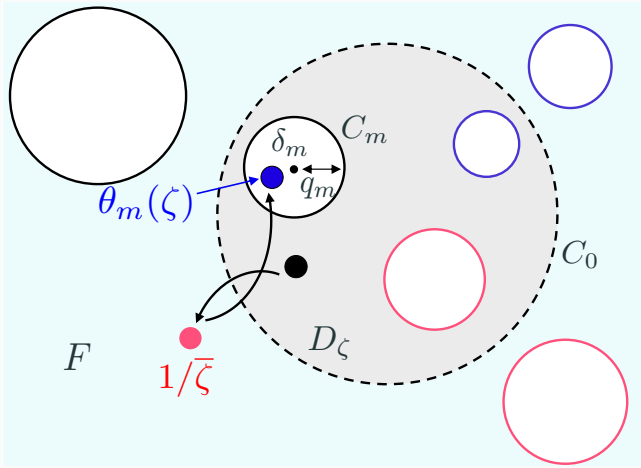
- Check that $a \mapsto 1/\overline{a}$ is a point reflected on the circle $|\zeta| = 1$
- Check that $a \mapsto \rho^2/\overline{a}$ is a point reflected on the circle $|\zeta| = \rho$
- What happens if you reflect a point on the circle $|\zeta - \delta_m| = q_m$?



One generator for each hole

An $(M+1)$-connected circular domain has hole centres δ_m and radii q_m. Inversion in circle m is

$$S_m(\zeta) = \overline{\delta_m} + \frac{q_m^2}{\zeta - \delta_m}.$$



That inversion, followed by inversion in the unit circle, is the Möbius map

$$\theta_m(\zeta) = \overline{S_m}(1/\zeta) = \delta_m + \frac{q_m^2 \zeta}{1 - \overline{\delta_m}\,\zeta}.$$

Inversion in C_m and then in C_n is $\theta_n \theta_m^{-1}$. $M = 2$ means three boundary circles.

The Schottky group

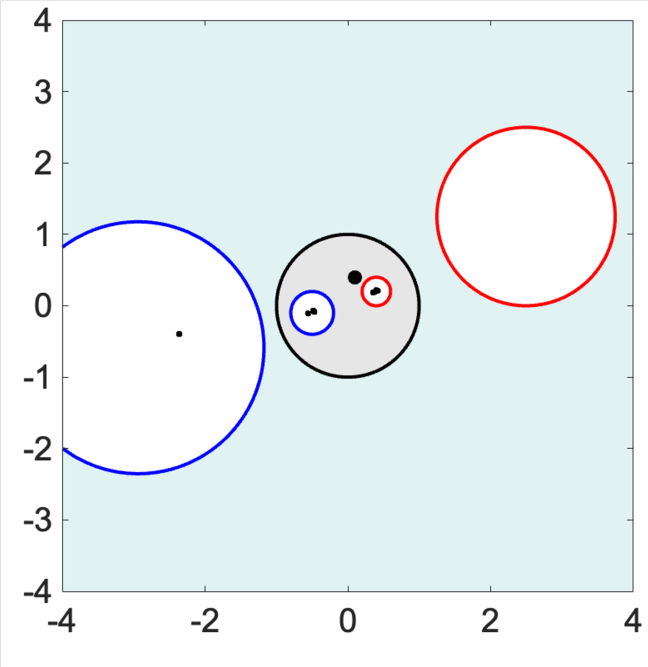
The $2M$ maps $\theta_m^{\pm 1}$ generate a free group Θ, the Schottky group of the circles:

$$\Theta = \{\theta_m, \theta_m^{-1} \mid m = 1, \ldots, M\}. \tag{9}$$

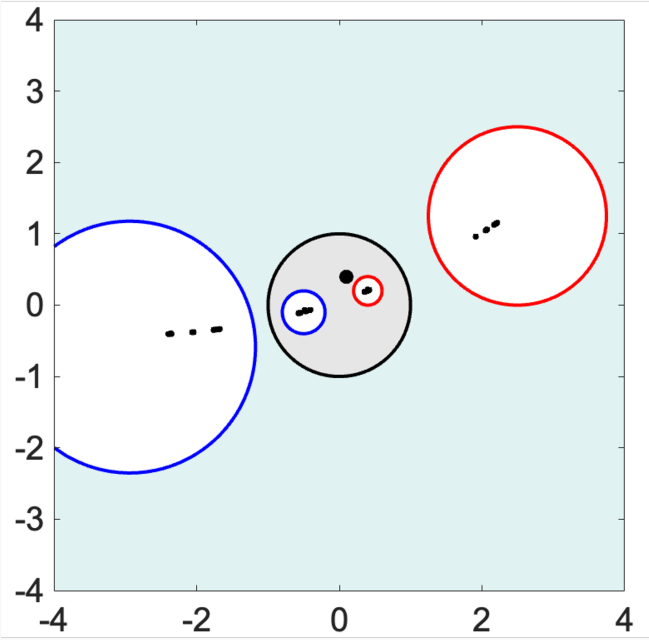
Every reflected copy of the vortex is $\theta(a)$ for some $\theta \in \Theta$.

Level 2

$\{\theta_1, \theta_2, \theta_1\theta_2, \theta_1^2, \theta_2^2, \theta_2\theta_1, \theta_1\theta_2^{-1}, \theta_1^{-1}\theta_2\}$



Level 6 (728 elements)



The prime function over the group

Drop the identity, and keep only one of θ and θ^{-1}. Call that set Θ'' (if $\theta_2\theta_3^{-1}$ is kept, $\theta_3\theta_2^{-1}$ is dropped).

Schottky–Klein prime functions (D. Crowdy, 2005)

$$\omega(\zeta, a) = (\zeta - a) \prod_{\theta \in \Theta''} \frac{(\theta(\zeta) - a)(\theta(a) - \zeta)}{(\theta(\zeta) - \zeta)(\theta(a) - a)}. \tag{10}$$

- The annulus is the special case. When $M = 1$, this product is Wednesday's $P(\zeta, \rho)$.
- This can be seen as a generalization of the annulus case.

Check that this is written as a product of the cross-ratio:

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}. \tag{11}$$

Schottky–Klein prime functions (D. Crowdy, 2005)

$$\omega(\zeta, a) = (\zeta - a) \prod_{\theta \in \Theta''} \frac{(\theta(\zeta) - a)(\theta(a) - \zeta)}{(\theta(\zeta) - \zeta)(\theta(a) - a)}. \tag{12}$$

- There is a single zero at $\zeta = \theta(a)$.
- There also has a zero at $\zeta = \theta^{-1}(a)$ because it satisfies $\theta(\zeta) = a$.

The vortex on every boundary circle

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{\omega(\zeta, \alpha)}{|\alpha|\, \omega(\zeta, 1/\overline{\alpha})} \right). \tag{13}$$

$\mathrm{Im}\, w$ is constant on each boundary circle, the same test as on the annulus.

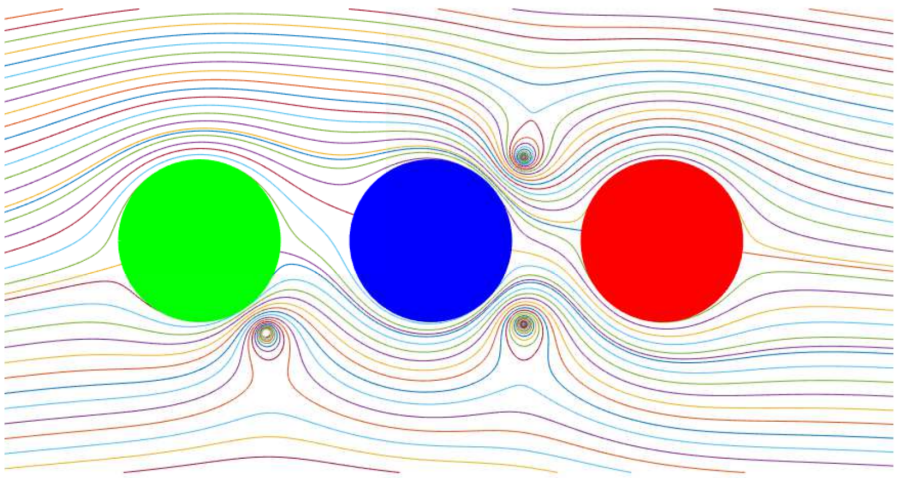


Figure 15.6. *Streamline distribution associated with uniform flow around three circular cylinders in the presence of three point vortices. There is no circulation around the cylinders. This figure was plotted using formulas (15.79)–(15.80).*

Crowdy, *Solving Problems in Multiply Connected Domains*, SIAM, 2020.

Conformal mapping

- Simply connected domain:

$$g(\zeta) = -\frac{\zeta - 1}{\zeta + 1} \tag{14}$$

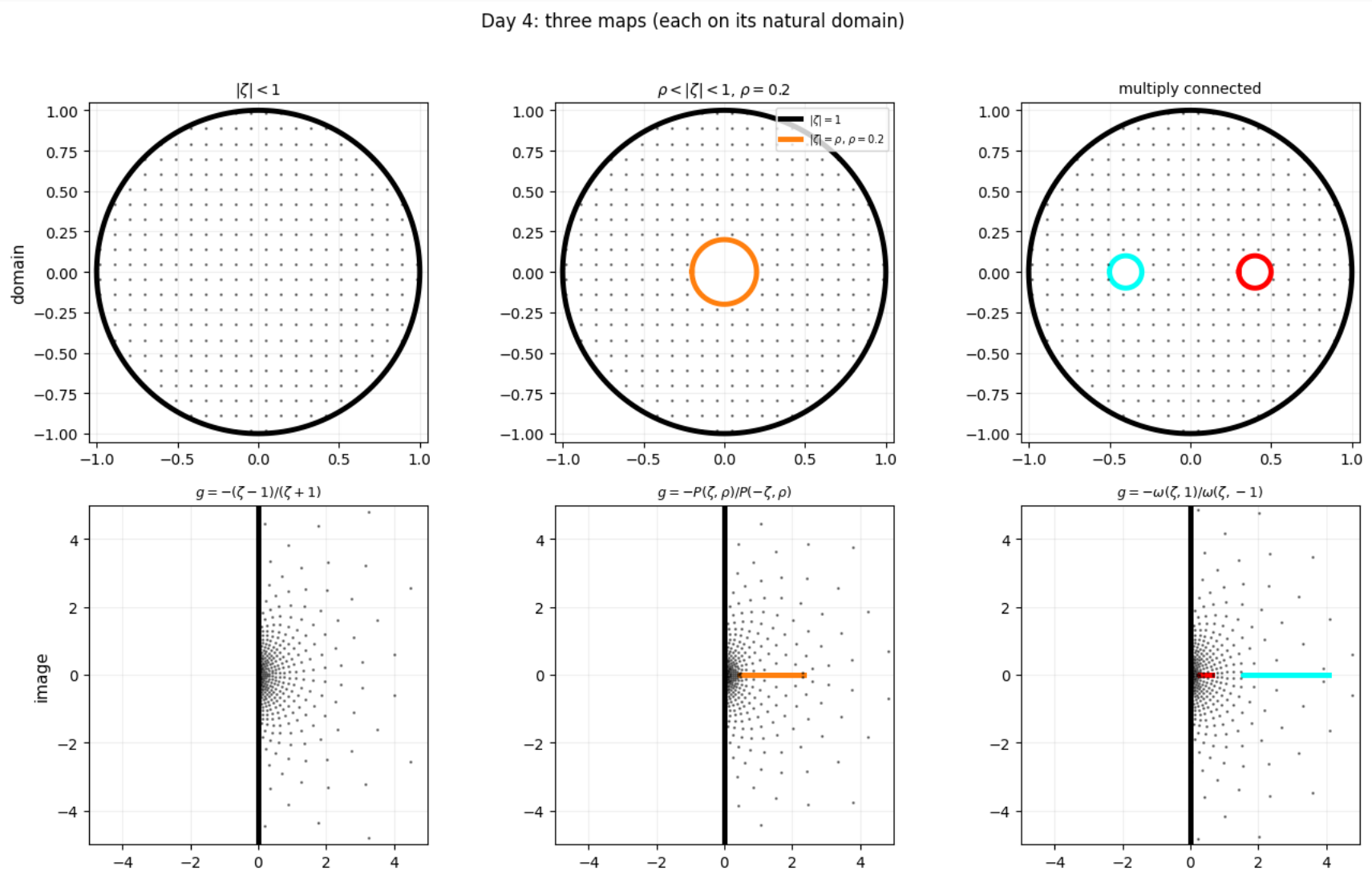
- Annulus:

$$g(\zeta) = \frac{P(\zeta, \rho)}{P(-\zeta, \rho)} \tag{15}$$

- Multiply connected domain:

$$g(\zeta) = -\frac{\omega(\zeta, 1)}{\omega(\zeta, -1)}. \tag{16}$$

Map


Day 4: three maps (each on its natural domain)
|ζ| < 1
ρ < |ζ| < 1, ρ = 0.2
multiply connected
|ζ| = 1
|ζ| = ρ, ρ = 0.2
domain
image
g = −(ζ − 1)/(ζ + 1)
g = −P(ζ, ρ)/P(−ζ, ρ)
g = −ω(ζ, 1)/ω(ζ, −1)

The product runs over words

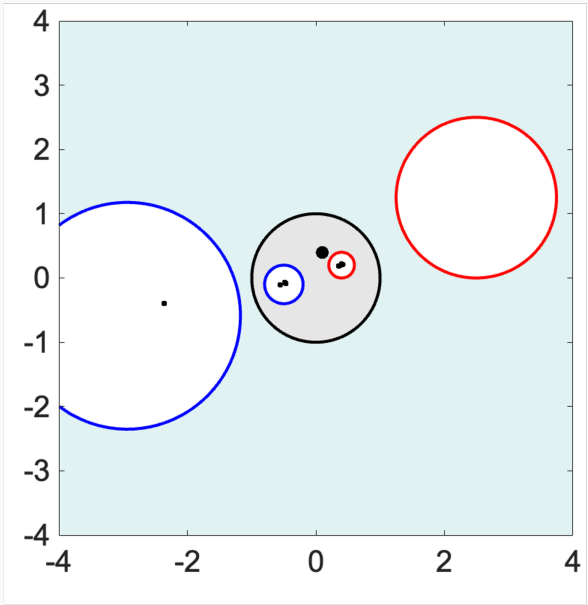
With M holes the generators do not commute. A reduced word of length L never cancels a generator against its own inverse, and there are

$$2M(2M-1)^{L-1} \tag{17}$$

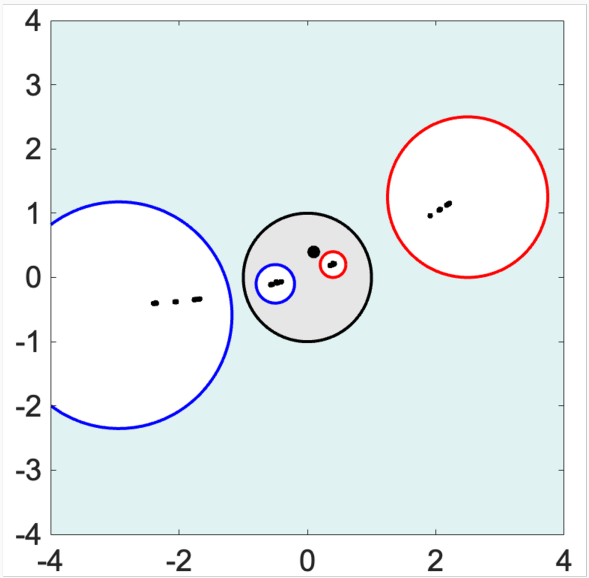
such words. The product for ω keeps one of θ and θ^{-1}, hence $M(2M-1)^{L-1}$ factors of length L. For two holes that sequence is $2, 6, 18, \ldots$, against two factors at each n on the annulus.

Level 2

$\{\theta_1, \theta_2, \theta_1\theta_2, \theta_1^2, \theta_2^2, \theta_2\theta_1, \theta_1\theta_2^{-1}, \theta_1^{-1}\theta_2\}$



Level 6 (728 elements)



One identity for each generator

Each generator acts on the prime function by a known factor (Crowdy, 2020):

$$\omega(\theta_m(\zeta), a) = \Gamma_m(\zeta, a)\,\omega(\zeta, a), \tag{18}$$

$$\Gamma_m(\zeta, a) = \sqrt{\theta_m'(\zeta)}\,\exp\Big(-2\pi\mathrm{i}\big(v_m(\zeta) - v_m(a) + \tfrac{1}{2}\tau_{mm}\big)\Big). \tag{19}$$

v_m is analytic off the holes and increases by 1 around the m-th hole. The period $\tau_{mm} = v_m(\theta_m(\zeta)) - v_m(\zeta)$ is a constant, fixed by the circles. The square root is the branch with $\omega(\zeta, a) \sim (\zeta - a)$.

On the annulus both generators have $\Gamma_m = -\zeta^{-1}$. That pair is FE1 and FE2.

Open problems

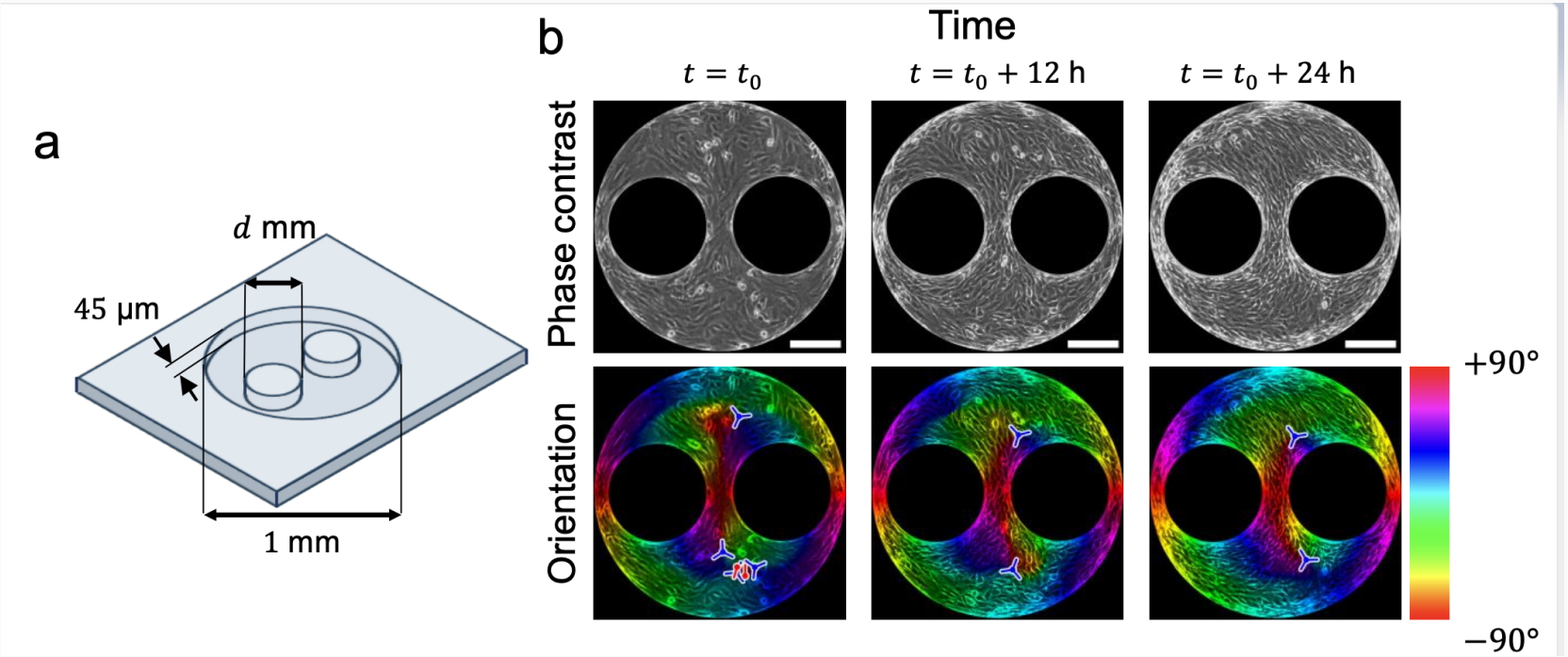
When the circles get close

- When the circles get close, the product $\omega(\zeta, a)$ becomes expensive to compute.
- We can truncate the product at a certain number of terms.
- This truncation is called the **truncation error**.
- The truncation error can be controlled by the graph theory [S. Yamamoto, H. Miyoshi, arXiv:2609.00785]

Application:
Cell culture experiments and multiply connected domains

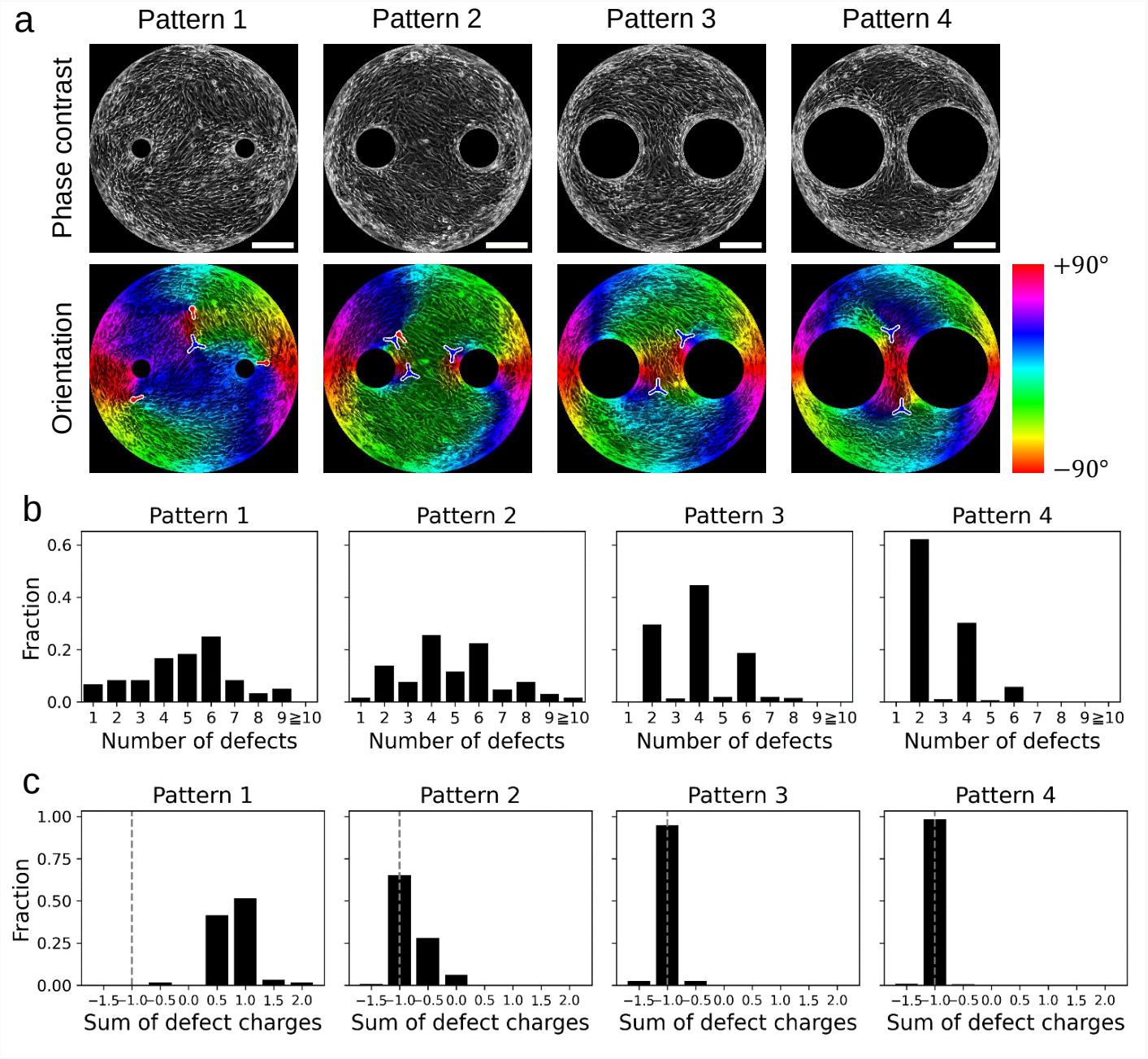
Two holes fix the total charge

- A $1\,\mathrm{mm}$ well with two pillars: triply connected, $\chi = 1 - h = -1$.
- Tangential anchoring forces the total defect charge to equal χ.
- Pairs annihilate. Two defects of charge $-1/2$ remain, and the mean charge stays -1.



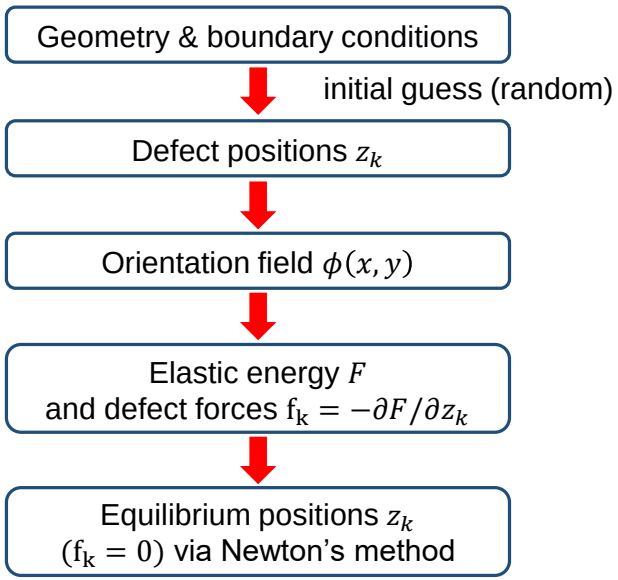
Kamao et al., arXiv:2609.03328, Fig. 1.

Large obstacles leave charge -1

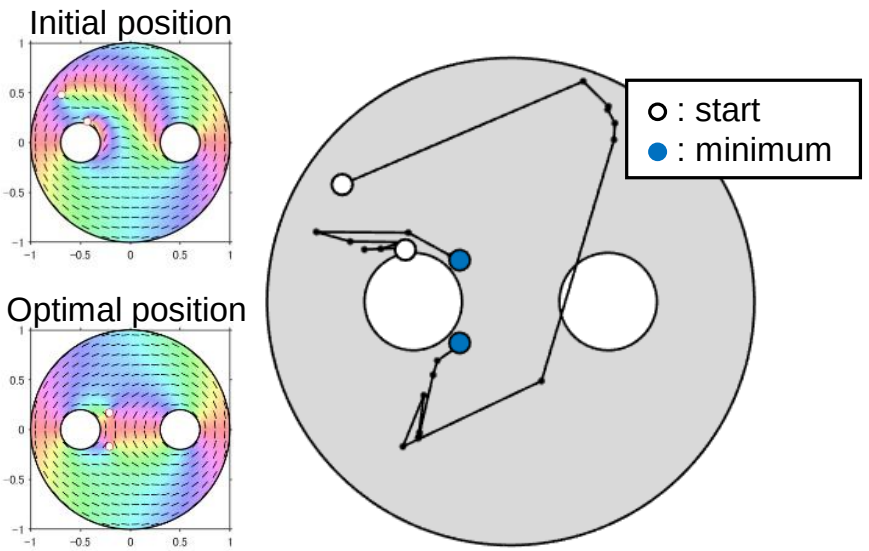


The orientation is a prime function

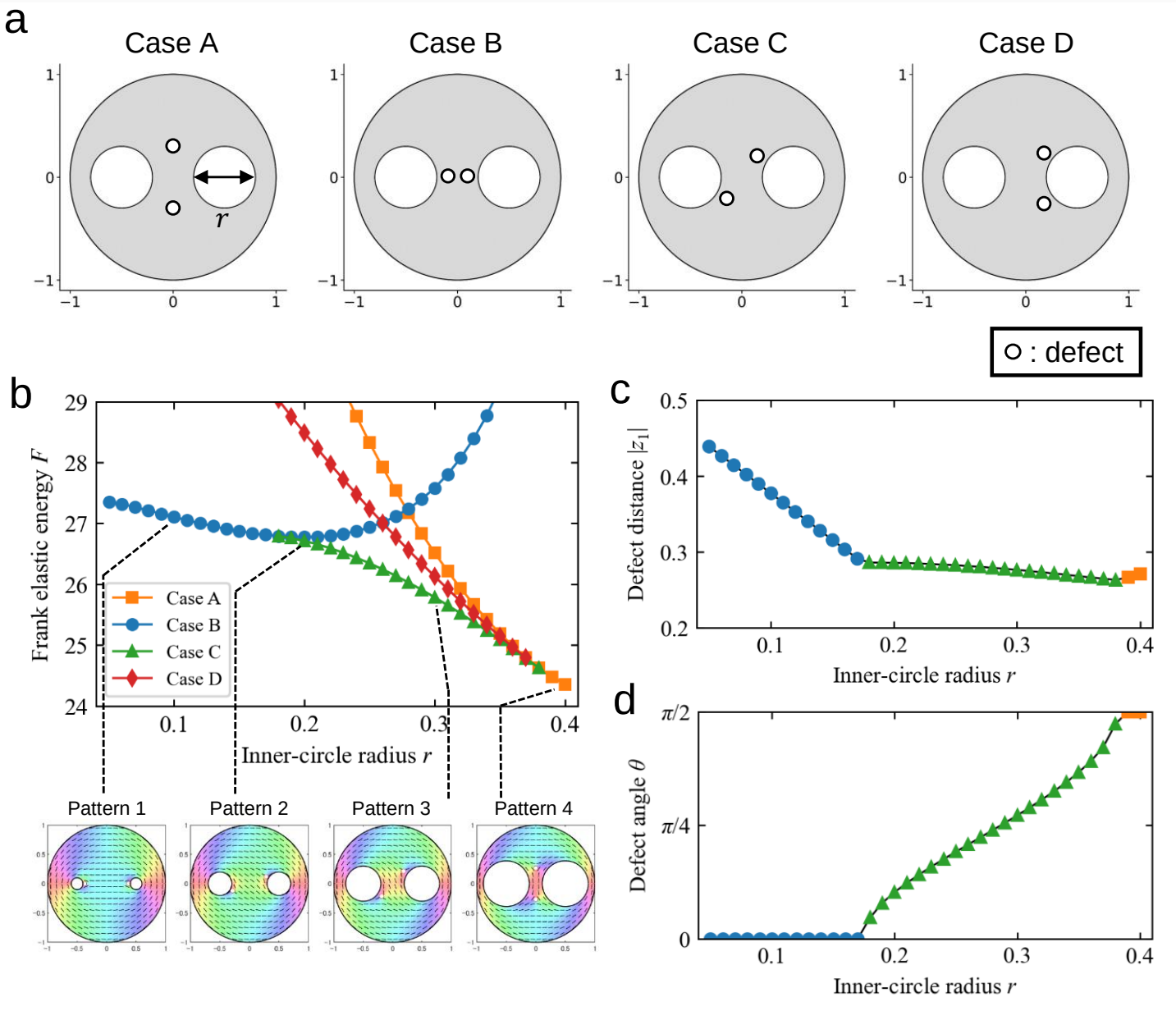
a



b



The radius selects the symmetry



Close

References

- `work/pinn_learning/03_schottky_klein_prime_nn.ipynb`
- Crowdy (2020), *Solving Problems in Multiply Connected Domains*, SIAM. The factor Γ_m for each Schottky generator.
- Raissi, Perdikaris, Karniadakis (2019), for the Monday residual loss
- Cuomo et al. (2022), *J. Sci. Comput.* **92**, 88: the loss, the initialization, and the open convergence question
- Farea, Yli-Harja, Emmert-Streib (2024), *AI* **5**(3), 1534–1557: a boundary constraint, checked by a test
- Kamao, Miyoshi, Nara, Miyazako (2026), arXiv:2609.03328. Two $-1/2$ defects in a triply connected cell monolayer.